

Addendum to Version 2.0

Logarithmic Perception Has a Physical Cause — and Its Degradation May Diagnose Alzheimer's Disease

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Five revisions following rigorous peer review.

Revision 1: Additional Citations

Three references were missing from Version 2.0 and should be added to the reference list and cited at the indicated locations in the text:

1a. Voytek & Knight (2015)

Voytek, B. & Knight, R.T. (2015). Age-Related Changes in 1/f Neural Electrophysiological Noise. *Journal of Neuroscience*, 35(38), 13257-13265.

This paper demonstrates, in two separate human studies, that 1/f electrophysiological noise increases with aging and that this age-related noise statistically mediates working memory decline. This is the most direct electrophysiological evidence for the neural noise hypothesis of aging upon which the biomarker prediction (Section 6) depends. It should be cited in Section 6.1 (Theoretical Basis) where the paper discusses additive noise accumulation during neurodegeneration, and in Section 6.2 (Supporting Evidence) as a fourth line of indirect evidence. Suggested insertion in Section 6.2: 'Fourth, Voytek and Knight (2015) demonstrated directly that 1/f neural noise increases with aging in human electrophysiological recordings, and that this noise increase statistically mediates age-related cognitive decline. Their results provide electrophysiological confirmation that the neural noise accumulation our framework invokes is a measurable, age-dependent phenomenon.'

1b. Ruderman (1994)

Ruderman, D.L. (1994). The statistics of natural images. *Network: Computation in Neural Systems*, 5(4), 517-548.

This paper provides early empirical measurements of natural image luminance statistics showing approximate log-normal distributions. It should be cited in Section 3.1 alongside the statement 'log-normal amplitude distributions are the predicted default statistics of natural environments, a

prediction well-supported by empirical measurements.' Suggested insertion: '...a prediction supported by empirical measurements of natural scene luminance distributions (Ruderman 1994; Field 1987).'

1c. Atick (1992)

Atick, J.J. (1992). Could information theory provide an ecological theory of sensory processing? Network: Computation in Neural Systems, 3(2), 213-251.

This paper is a foundational argument for information-theoretic efficient coding of natural statistics as an explanation for sensory neural response properties. It should be cited in Section 2.3 alongside Barlow (1961) and Laughlin (1981). Suggested insertion in Section 2.3: 'Atick (1992) formalized this program, asking whether information theory could provide a complete ecological theory of sensory processing.'

Revision 2: Abstract Language Corrections

Two phrases in the abstract of Version 2.0 should be softened to accurately reflect the status of the biomarker claim:

Change 1. The abstract currently states: '...the Stevens exponent for luminance provides a non-invasive readout of additive neural noise accumulation: as neurons die in Alzheimer's disease, the exponent should drift measurably upward, detectable with a 10-minute brightness-scaling task years before cognitive symptoms appear.'

This should read: '...the Stevens exponent for luminance may provide a non-invasive readout of additive neural noise accumulation: as neurons die in Alzheimer's disease, the exponent is predicted to drift measurably upward, potentially detectable with a 10-minute brightness-scaling task years before cognitive symptoms appear. This prediction has not yet been empirically tested.'

Change 2. The Clinical Significance box on page 1 states: 'This paper proposes a fundamentally different approach: a 10-minute computerized brightness test, requiring only a standard display and number pad, grounded in a physical theory of how the brain encodes sensory information.'

This should add at the end: 'The approach is currently a theoretical prediction requiring prospective empirical validation; no clinical data are presented in this paper.'

Revision 3: Section 5 Title Change

The title of Section 5 currently reads: 'Dark Adaptation as an SNR Crossover.'

This should be changed to: 'Dark Adaptation Reinterpreted as a Possible SNR Crossover.'

The section content already uses appropriate qualifying language ('We present this as a qualitative prediction only'), but the title should match this level of caution.

Revision 4: New Discussion Subsection — Consequences of Noise Injection Site Failure

The following subsection should be inserted after Section 8.2 (Explicit Statement of Assumptions) as Section 8.2a:

8.2a. Consequences if the Noise Injection Site Assumption Fails

The derivation in Section 3.3 requires that neural noise enters the sensory circuit predominantly after the logarithmic transduction stage. For vision, this is supported by known retinal architecture: outer retinal circuits (photoreceptors and horizontal cells) perform logarithmic gain control, while the primary sources of stochastic neural noise (synaptic vesicle release variability, ion channel noise, cortical background activity) accumulate in the inner retina and downstream visual cortex.

For other modalities, the noise injection site is less well-characterized. In audition, the cochlear amplifier (outer hair cells) provides nonlinear compression before the inner hair cells transduce mechanical vibration into neural signals. If significant noise is introduced at the outer hair cell stage (before compression), the multiplicative noise derivation would not hold for audition, and the predicted relationship between N_{mult} and the auditory Stevens exponent would require a different explanation.

For somatosensation, olfaction, gustation, and pain, the transduction and noise architecture are less thoroughly mapped. The framework's predictions for these modalities should be treated as contingent on future verification of their noise injection site.

If the noise injection site assumption fails for a specific modality, three consequences follow: (1) the multiplicative noise derivation (Section 3.3) does not apply for that modality, though the multiplicative CLT argument for log-normal stimulus distributions (Section 3.1) remains valid; (2) the N_{mult} ordinal prediction (Table 1) loses that modality as a data point; and (3) the neurodegeneration biomarker prediction (Section 6) applies only to modalities where the assumption holds. The biomarker hypothesis as proposed targets brightness (vision), where the assumption is best supported. It does not depend on the assumption holding for all modalities.

Revision 5: New Discussion Subsection — Year-to-Year Measurement Variability as the Primary Practical Barrier

The following subsection should be inserted after Section 8.3 (Limitations) as Section 8.3a:

8.3a. Year-to-Year Exponent Variability: The Critical Practical Barrier

The feasibility of the biomarker depends entirely on whether within-individual year-to-year variability in measured Stevens exponents is small enough that pathological drift can be distinguished from measurement noise. Teghtsoonian and Teghtsoonian (1983) found that Stevens exponents are reliable over 24-hour intervals (significant test-retest correlation) but not over one-year intervals (non-significant correlation). We reinterpret this finding in Section 6.2 as potential evidence that slow neural drift contributes to the year-long variability, but this interpretation is speculative.

The conservative interpretation is that the one-year test-retest standard deviation of the Stevens exponent for brightness is on the order of 0.05 to 0.10 in healthy adults. If the pathological drift rate in preclinical Alzheimer's disease is also 0.05 per year, as we provisionally suggest as a referral threshold, then the signal-to-noise ratio for single-year comparisons is approximately 0.5 to 1.0 — marginal at best.

Three design features of the proposed protocol address this challenge. First, the biomarker criterion is trajectory over multiple years, not a single-year comparison. A sustained upward

slope across three or more annual measurements is far less likely to arise from random measurement variability than a single elevated value. Second, the within-session protocol (25-30 luminance levels, magnitude estimation) can be optimized for test-retest reliability by standardizing instructions, display calibration, ambient lighting, and response modality. Third, the specificity control (simultaneous measurement of warmth or pain exponents) provides an internal check: if year-to-year variability is driven by non-specific factors (attention, motivation, strategy), it should appear equally in all modalities, not selectively in brightness.

Nevertheless, the year-to-year variability of the Stevens exponent remains the single most important practical barrier to the proposed biomarker. A dedicated test-retest reliability study — measuring brightness Stevens exponents in 50+ healthy adults at 1-month, 6-month, and 12-month intervals under standardized conditions — is a necessary prerequisite to any clinical validation trial. If the 12-month test-retest standard deviation exceeds 0.10 in healthy adults, the biomarker as currently proposed would require either substantially longer observation windows or improved measurement protocols to achieve adequate sensitivity.

Revision 6: N_mult Correlation Caveat in Abstract

The abstract currently states: 'First, we propose that Stevens exponents should decrease monotonically with the number of independent multiplicative factors N_mult in each modality's stimulus physics — an ordinal prediction supported by Spearman rank correlation across eight modalities ($r_s = -0.81$, $p = 0.015$).'

This should be amended to: 'First, we propose that Stevens exponents should decrease monotonically with the number of independent multiplicative factors N_mult in each modality's stimulus physics — a preliminary ordinal prediction supported by Spearman rank correlation across eight modalities ($r_s = -0.81$, $p = 0.015$, $N = 8$) with hand-assigned N_mult values. Pearson correlation does not reach significance ($r = 0.57$, $p = 0.09$). Confirmation requires preregistered N_mult assignments across 15 or more modalities (Prediction 1).'

Summary of All Changes

1. Three citations added: Voytek & Knight (2015), Ruderman (1994), Atick (1992).

2. Abstract softened: 'provides' changed to 'may provide'; 'detectable' changed to 'potentially detectable'; clinical significance box adds disclaimer.

3. Section 5 title: 'as an SNR Crossover' changed to 'as a Possible SNR Crossover.'

4. New Section 8.2a: discusses consequences if noise injection site assumption fails for non-visual modalities.

5. New Section 8.3a: discusses year-to-year measurement variability as the primary practical barrier to the biomarker.

6. Abstract N_{mult} correlation: adds 'preliminary,' notes non-significant Pearson, points to Prediction 1 for confirmation.

No changes were made to the mathematical derivations, the core theoretical framework, or the experimental protocol. All revisions address presentation, scope, and completeness of discussion.